Advanced techniques in applied economics

Lecture 3: DAGs, interventions, and structural equation models

Piotr Zwiernik

Universitat Pompeu Fabra Barcelona

Statistics, Probability and Machine Learning Research Group

Barcelona School of Economics

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From undirected structure to causal structure

What changes today?

Today we move to **directed** graphs, where arrows encode asymmetric dependence and can be used to describe structural mechanisms.

Undirected graphs
Useful for summarizing conditional dependence networks.
They are symmetric and mainly descriptive.

Directed acyclic graphs
Useful for expressing directional or causal assumptions about how variables are generated.

Main question: Which variables matter for which others, and how would the system react to intervention?

Roadmap for today

1. DAGs and factorization
2. d-separation: chain, fork, collider
3. moralization as a way to test d-separation
4. Markov equivalence and CPDAGs
5. interventions and mechanism changes
6. linear SEMs and Wright's path tracing rules

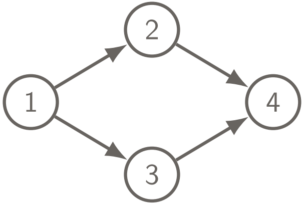
Memorable statement: a DAG is both a probabilistic model and a structural story.

Part 1: DAGs as structural models

What is a DAG?

A **directed acyclic graph** (DAG) consists of

- vertices $1, \ldots, m$ representing variables $X_1, \ldots, X_m$,
- directed edges $i \to j$,
- no directed cycles.



Interpretation

The arrow $i \to j$ says that X_i is a direct input into the mechanism generating X_j.

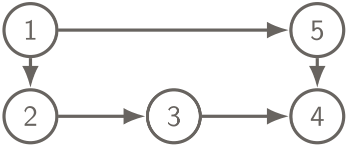
For standard definitions see Chapter 3 in Højsgaard, Edwards, Lauritzen, Graphical models in R.

Parents, children, ancestors, descendants

Let D be a DAG.

For a node j:

- $\mathrm{pa}(j)$ = parents of j,
- $\mathrm{ch}(j)$ = children of j.
- $\mathrm{an}(j)$ = ancestors of j,
- $\mathrm{de}(j)$ = descendants of j.



For node 4:

$\mathrm{pa}(4) = \{3, 5\}$, $\mathrm{an}(4) = \{1, 2, 3, 5\}$.

For node 1:

$\mathrm{ch}(1) = \{2, 5\}$, $\mathrm{de}(1) = \{2, 3, 4, 5\}$.

Factorization over a DAG

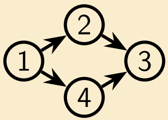
A distribution P is **Markov with respect to a DAG** D if its density factorizes as

$$p(x_1, \ldots, x_m) = \prod_{j=1}^{m} p\big(x_j \mid x_{\mathrm{pa}(j)}\big).$$

In a DAG, each factor involves one variable and its parents.

As in the undirected case, the graph has a dual role:

- it determines a factorization of the joint law,
- it implies conditional independence statements.



Then the joint law factorizes as:
$p(x_1, x_2, x_3, x_4) = p(x_1)\, p(x_2 \mid x_1)\, p(x_4 \mid x_1)\, p(x_3 \mid x_2, x_4)$.

For DAG factorization and the Markov property, see Lauritzen, *Graphical Models*, Chapter 3.

From DAGs to structural equations

A DAG can be viewed not only as a factorization of a joint law, but also as a system of structural assignments.

Structural equation view

For each node j, write

$$X_j = f_j\big(X_{\mathrm{pa}(j)}, \varepsilon_j\big), \qquad j = 1, \ldots, m,$$

where the noise variables $\varepsilon_1, \ldots, \varepsilon_m$ are jointly independent.

Why this matches the DAG

Each variable depends only on its parents and on its own noise term. So the arrows describe which variables enter directly into each mechanism.

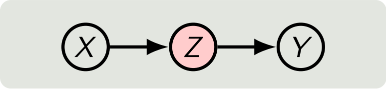
For the structural equation interpretation of DAGs, see Pearl, *Causality*, Chapter 1, or Peters, Janzing, and Schlkopf, *Elements of Causal Inference*, Chapter 5.

Part 2: d-separation

Three basic patterns

Everything starts with three local configurations.

Chain

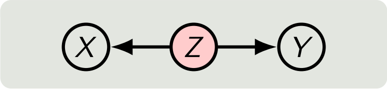


Always:
$p(x, y, z) = p(x)\, p(z \mid x)\, p(y \mid x, z)$

Here:
$p(x, y, z) = p(x)\, p(z \mid x)\, p(y \mid z)$

$$X \perp\!\!\!\perp Y \mid Z$$

Fork

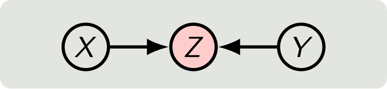


Always:
$p(x, y, z) = p(z)\, p(x \mid z)\, p(y \mid x, z)$

Here:
$p(x, y, z) = p(z)\, p(x \mid z)\, p(y \mid z)$

$$X \perp\!\!\!\perp Y \mid Z$$

Collider



Always:
$p(x, y, z) = p(x)\, p(y \mid x)\, p(z \mid x, y)$

Here:
$p(x, y, z) = p(x)\, p(y)\, p(z \mid x, y)$

$$X \perp\!\!\!\perp Y$$

Two important notes

- Conditioning **blocks** a chain or fork, but **opens** a collider.
- Different DAGs may define the same model!

The chain pattern $X \perp\!\!\!\perp Y | Z$

$$X \longrightarrow Z \longrightarrow Y$$

Information flowing from X to Y

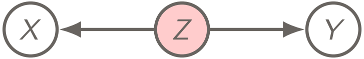
Information flows from X to Y unless we condition on Z blocking the path.

We say that the path is **open** if we do not condition on Z.

Policy example
Policy X changes investment Z, which changes output Y. After fixing investment, policy adds no further predictive content for output.

Finance example
Monetary news X moves interest rates Z, which then affect stock returns Y. After fixing interest rates, monetary news adds no further predictive content for returns.

The fork pattern



Information flowing from X to Y

Information flows from X to Y unless we condition on Z blocking the path.

We say that the path is **active** if we do not condition on Z.

Demand-shock example
A common demand shock Z affects both price X and quantity Y. After controlling for the shock, the induced association disappears.

Ability example
Ability Z affects both education X and earnings Y. Conditioning on ability removes the spurious component of the association.

The collider pattern

Information flowing from X to Y

Information flows from X to Y only if we condition on Z opening the path.

We say that the path is **active** if we condition on Z.

Selection example
Admission Z to a program may depend on both test scores X and extracurricular achievements Y. Among admitted students, these may appear negatively related.

Labor-market example
Remaining in a demanding occupation Z may depend on both ability X and motivation Y. Conditioning on those who remain can induce dependence between them.

d-separation: the general rule

Conditional independence is not encoded by a simple separation but rather **d-separation**.

A path is **active given S** if

- every non-collider on the path is outside S,
- every collider on the path has itself or one of its descendants in S.

Equivalently, a path is **blocked given S** if it is not active given S.

d-separation and conditional independence

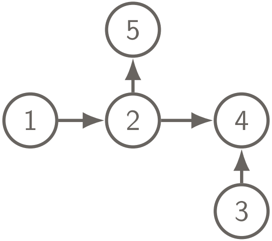
Two sets A and B are **d-separated by S** if every path between A and B is blocked given S.

If A and B are d-separated by S in the DAG, then $X_A \perp\!\!\!\perp X_B \mid X_S$.

For d-separation, see e.g. Pearl, *Causality*, Section 1.2.3.

A longer example

Consider the DAG



- 1 and 3 are d-separated by $\emptyset$ because the only path hits the collider 4.
- 1 and 3 become d-connected if we condition on 4.
- 1 and 4 are d-separated by $\{2\}$ because the chain is blocked at 2.

Moralization: a practical way to test d-separation

A useful way to test d-separation is to convert the DAG into an undirected graph.

1. To test $X_A \perp\!\!\!\perp X_B | X_S$ keep only the ancestors of $A \cup B \cup S$.
2. **Moralize**: connect any two parents of a common child.
3. Drop all arrow directions.
4. Check whether S separates A and B in the resulting undirected graph.

Moralization criterion

A and B are d-separated by S in the DAG if and only if S separates A and B in the moralized ancestral graph.

For moralization and ancestral graphs in DAG separation, see Lauritzen, *Graphical Models*, Chapter 3.

Part 3: Markov equivalence and CPDAGs

Markov equivalence: why observational data do not identify every arrow

Different DAGs can encode the same set of conditional independence relations.

DAGs are **Markov equivalent** if they imply the same conditional independence statements.

Example

The following three DAGs are Markov equivalent. All of them imply $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

$$1 \rightarrow 2 \rightarrow 3 \qquad 1 \leftarrow 2 \leftarrow 3 \qquad 1 \leftarrow 2 \rightarrow 3$$

The collider $1 \rightarrow 2 \leftarrow 3$ is **not** equivalent to the three above.

From observational data, we can typically recover only a Markov equivalence class.

Exact DAG identification may be possible under additional assumptions.

The Verma–Pearl characterization and CPDAGs

A **v-structure** is a triple of the form $X \to Z \leftarrow Y$ with no edge between X and Y.

Theorem (Verma–Pearl)

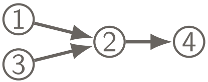
Two DAGs are Markov equivalent if and only if they have

- the same **skeleton**,
- the same **v-structures**.

A **CPDAG** represents an entire Markov equivalence class.

- **Directed edges:** directions shared by every DAG in the class.
- **Undirected edges:** directions that cannot be identified from observational data alone.

All v-structures are fixed across the equivalence class, but there may be more fixed arrows.



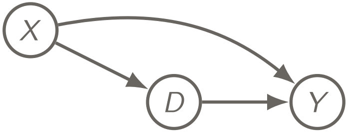
Above $1 \to 2 \leftarrow 3$ is a v-structure, but the arrow $2 \to 4$ is also fixed.

For the characterization of Markov equivalence, see Verma and Pearl (1990) or Chickering (1995).

Part 4: Interventions and policy changes

Why observational comparisons can be misleading

Suppose we want the causal effect of D on Y.



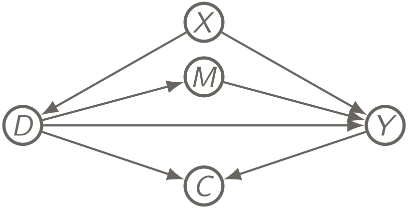
There is a causal path $D \rightarrow Y$ but also a noncausal path $D \leftarrow X \rightarrow Y$.
Because of this, the observational association between D and Y mixes

- the effect of D on Y,
- and the confounding created by X.

A path such as $D \leftarrow X \rightarrow Y$ is called a **back-door path** because it enters the treatment node through an incoming arrow.

What should we condition on?

A realistic causal graph usually contains several observed variables, but not all of them are good controls.



Adjust for X	Do not adjust for M	Do not adjust for C
X is a pre-treatment confounder. It blocks the back-door path $D \leftarrow X \rightarrow Y$.	M is a mediator. Conditioning on it blocks part of the causal effect of D on Y.	C is a collider. Conditioning on it opens a spurious path between D and Y.

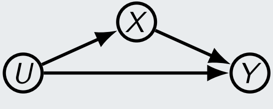
Back-door adjustment is not control for everything. It is control for the right pre-treatment confounders, but not mediators or colliders.

Observing is not the same as intervening

Observational data tell us how variables move together. Policy analysis asks a different question:

What happens if we actively change one mechanism?

Simple setup



X = policy variable
Y = economic outcome
U = background conditions

Observational question
What is $\mathbb{P}(Y \mid X = x)$?
This is the outcome distribution among units with observed value $X = x$.

Interventional question
What is $\mathbb{P}(Y \mid \mathrm{do}(X = x))$?
This is the outcome distribution if we actively set X to x.

If U affects both X and Y, then typically $\mathbb{P}(Y \mid X = x) \neq \mathbb{P}(Y \mid \mathrm{do}(X = x))$.

For the do-operator and interventional distributions, see Pearl, *Causality*, Chapter 3.

Hard interventions: the *do* operator

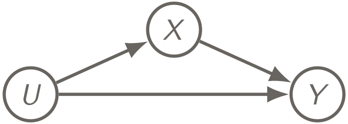
A hard intervention replaces the mechanism generating one variable by an external assignment:

$$\mathrm{do}(X = x).$$

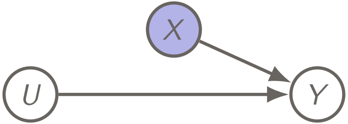
Graphical meaning

A hard intervention removes all arrows pointing into the intervened variable. The original mechanism for X is replaced by an external assignment.

Observational regime



After $\mathrm{do}(X = x)$



Because interventions change the graph in a controlled way, they can be used to recover causal directions that are not identifiable from observational data alone.

A tiny intervention example in R

Suppose the true system is

$$X \to Y, \qquad Y = X + \varepsilon_Y.$$

Observationally, association alone does not tell us whether X causes Y or vice versa. But under intervention on X, the response of Y becomes directly visible.

```r
n <- 1000
# observational regime
x.obs <- rnorm(n)
y.obs <- x.obs + rnorm(n, sd = 0.5)
# intervention do(X = 2)
x.int <- rep(2, n)
```

```r
y.int <- x.int + rnorm(n, sd = 0.5)

mean(y.obs)
mean(y.int)
plot(x.obs, y.obs, pch = 16, cex = .4)
abline(v = 2, col = 2, lwd = 2)
```

Interpretation

Under do($X = 2$), the distribution of Y shifts. This reveals the direction $X \to Y$.

We could do the same with **soft interventions**, where the mechanism changes without fixing a variable to a constant. Think about how such interventions could also help orient a DAG.

Mechanism stability across environments

Not always interventions are possible. Suppose we observe data from **several environments**.

True direction: $X = f(\varepsilon_X), \qquad Y = g(X, \varepsilon_Y).$

Environment change Suppose the distribution of X changes across environments, for example because the mechanism for X changes ($X = f^{(e)}(\varepsilon_X)$) but the mechanism from X to Y stays the same ($Y = g(X, \varepsilon_Y)$).

Key implication

Across environments, $P^{(e)}(X)$ may change, but the conditional law $P^{(e)}(Y \mid X)$ stays stable in the correct causal direction.

This is not a new intervention concept. It is an identification idea: the right causal direction is the one with the stable mechanism.

What changes and what stays the same?

Suppose the true direction is $X \to Y$ and the mechanism for X changes across environments.

What may change?
Because the distribution of the input changes, we generally have

$$P^\star(X) \neq P(X), \qquad P^\star(Y) \neq P(Y).$$

Usually the reverse conditional changes as well:

$$P^\star(X \mid Y) \neq P(X \mid Y).$$

What stays stable?
If the mechanism from X to Y is unchanged, then

$$P^\star(Y \mid X) = P(Y \mid X).$$

Invariance-based discovery looks for a direction in which the conditional mechanism remains stable when environments change.

For invariance ideas in causal discovery, see Peters, Bühlmann, and Meinshausen (2016), *JRSS-B*, and Peters, Janzing, and Schlkopf, *Elements of Causal Inference*, Chapter 13.

A modern economics example: climate and weather

Basic idea: Use short-run weather variation across regions and years as different environments, and ask whether the response of economic outcomes to these shocks is stable.

Examples of outcomes

- agricultural output,
- regional income,
- productivity,

Examples of environments

- different regions,
- different years,
- different weather realizations,

The key identifying hope is that the causal response mechanism is stable across environments, even though the distribution of shocks changes.

For policy invariance in economics, see Heckman (1999, 2008). For a modern climate-econometrics example, see Lemoine (2018), *Estimating the Consequences of Climate Change from Weather Variability*.

Part 5: Linear SEMs and path tracing

Linear structural equation models

A linear SEM associated with a DAG has the form

$$X_i = \sum_{j \in \mathrm{pa}(i)} \beta_{ij} X_j + \varepsilon_i, \qquad i = 1, \dots, m,$$

where $\varepsilon_1, \dots, \varepsilon_m$ are mutually independent (or at least satisfy $\mathbb{E}(\varepsilon_i \mid X_{\mathrm{pa}(i)}) = 0$).

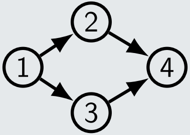
Matrix form: $X = BX + \varepsilon,$

where $B_{ij} = 0$ when $j \notin \mathrm{pa}(i)$.

Note: The zero pattern of B is exactly the DAG.

In the example, X_1 is a primitive shock, X_2 and X_3 are intermediate channels, and X_4 is the outcome of interest.

$$\begin{aligned} X_1 &= \varepsilon_1, \\ X_2 &= \beta_{21} X_1 + \varepsilon_2, \\ X_3 &= \beta_{31} X_1 + \varepsilon_3, \\ X_4 &= \beta_{42} X_2 + \beta_{43} X_3 + \varepsilon_4. \end{aligned}$$



For linear SEMs, see Bollen, *Structural Equations with Latent Variables*, Chapter 3, or Drton, Eichler, and Richardson in the *Handbook of Graphical Models*.

Gaussian LSEMs are exactly Gaussian DAG models

Recall: a DAG model is defined by the factorization $f(x) = \prod_{i=1}^{m} f(x_i \mid x_{\mathrm{pa}(i)})$.

If X is jointly Gaussian, then each conditional distribution $X_i \mid X_{\mathrm{pa}(i)}$ is Gaussian. So a Gaussian DAG model is specified by choosing these Gaussian conditionals.

Gaussian LSEM writes

$$X_i = \sum_{j \in \mathrm{pa}(i)} \beta_{ij} X_j + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \omega_i),$$

with independent Gaussian errors.

Conditional law:

$$X_i \mid X_{\mathrm{pa}(i)} \sim N\left(\sum_{j \in \mathrm{pa}(i)} \beta_{ij} X_j, \omega_i\right).$$

Gaussian LSEMs are exactly Gaussian DAG models

Comparing the two, we conclude that linear Gaussian SEM specifies exactly the Gaussian conditionals appearing in the DAG factorization.

Structural coefficients and total effects

Structural coefficients
In the linear SEM $X = BX + \varepsilon$, B records the **direct** linear effects encoded by the DAG.

Reduced-form map Solving the system gives

$$X = (I - B)^{-1}\varepsilon.$$

So the matrix $(I - B)^{-1}$ combines **direct and indirect** effects.

Structural coefficients describe local mechanisms.
The reduced-form map describes the total propagation of shocks.

Wright's path tracing: the basic idea

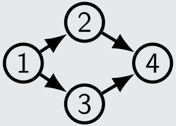
In a **linear** SEM, the covariance between two variables can be written as contributions from different paths in the graph.

This helps explain an association as a combination of

- direct effects,
- indirect effects through mediators,
- common-cause effects.

$$X_2 = \beta_{21}X_1 + \varepsilon_2,$$
$$X_3 = \beta_{31}X_1 + \varepsilon_3,$$
$$X_4 = \beta_{42}X_2 + \beta_{43}X_3 + \varepsilon_4,$$

with $\mathrm{var}(X_1) = 1$.



Path tracing calculation: To understand $\mathrm{cov}(X_1, X_4)$, we look at the two directed paths $1 \to 2 \to 4$ and $1 \to 3 \to 4$. Each path contributes the product of its coefficients, so $\mathrm{cov}(X_1, X_4) = \beta_{21}\beta_{42} + \beta_{31}\beta_{43}$, assuming $\mathrm{var}(X_1) = 1$.

For Wright's path tracing rules, see Wright (1934).

Take-away

- A DAG factorizes the joint law into local conditional mechanisms.
- d-separation tells us which conditional independences the DAG implies.
- Different DAGs may be Markov equivalent, so observational data do not identify every arrow.
- Interventions modify mechanisms, so $\mathbb{P}(Y \mid X = x)$ and $\mathbb{P}(Y \mid \mathrm{do}(X = x))$ are different objects.
- In linear SEMs, the graph appears directly in the coefficient matrix, and path tracing explains how effects propagate.

Main lesson of today: a DAG is both a statistical model for conditional independence and a structural model for interventions.

Looking ahead

Next lecture

We s move from **representing** causal structure to **learning** it from data.

What comes next?

- constraint-based learning from conditional independences,
- score-based and structural approaches,
- when observational data identify only an equivalence class,
- how extra assumptions such as non-Gaussianity can help orient edges.

Today the DAG was assumed known. Next time the question becomes: how much of it can we recover from data?